

Screen Sense: Kids' Screen-time Visualization

Milestone 3:

Segment & Insight Deep-Dives

Week 5: Cohort and Segment Analysis

- Identify top cohorts (e.g., age bands × device types)
- Heatmaps/stacked comparisons by demographic or location segments

Week 6: Seasonal/Calendar or Habit Patterns (if applicable)

- Monthly or term-time comparisons (if dates exist)
- Summarize segment-wise insights and possible drivers

Deliverables: Seasonal/segment summaries and cohort insights.

Objective of Milestone 3:

The objective of Milestone 3 is to move beyond basic visual exploration and deeply analyze **behavioral segments, usage cohorts, and time-based patterns** within the dataset. This milestone focuses on identifying **which groups of children exhibit the highest screen-time exposure**, how screen-time patterns differ across demographics, and how habits vary between weekdays and weekends or across seasonal periods. By performing cohort comparisons, heatmaps, stacked segmentation views, and habit trend analysis, this milestone generates **actionable insights** that help explain the drivers behind screen-time behavior among different age groups, device users, and location segments.

WEEK 5: Cohort and Segment Analysis:

Week 5 focuses on grouping children into **cohorts** based on demographic and behavioral attributes, then analyzing each segment's screen-time patterns.

1. Identifying Top Cohorts (Age Band × Device Type)

In this step, children are grouped by their **age band** (such as 8–11, 12–15, 16–18) and **primary device** (smartphone, laptop, TV, tablet).

Average screen time is calculated for each combination to identify:

- Which age band has the highest device usage
- Which device type contributes most to excessive screen time
- How usage patterns evolve as children grow older

This helps discover segments such as:

"Urban Teens using Smartphones" → Highest-risk cohort.

2. Heatmaps for Multi-Dimensional Insights

Heatmaps are used to visualize the intensity of screen-time usage across segments such as:

- **Age Band × Device Type**
- **Age Band × Location (Urban/Rural)**
- **Device Usage × Gender**

Heatmaps allow quick detection of high-exposure clusters, showing visually where screen-time is concentrated.

Example findings:

- Smartphone usage is dominant for the highest screen-time bands.
- Urban regions show consistently higher device usage intensity

3. Stacked Comparisons by Demographic Segments

Stacked bar charts compare the distribution of screen-time levels or device types across groups such as:

- Gender
- Urban vs Rural
- Age groups

These comparisons highlight:

- Which demographic groups use which devices more frequently
- Which groups maintain balanced vs excessive screen-time behavior
- Whether location influences technology access and screen habits

For example:

Urban children tend to use mobile devices and laptops more, while rural children rely more on TVs or shared devices.

4. Multi-Dimensional Cohort Comparison (Age × Device × Location)

A 3-level cohort heatmap was created to explore combined effects of:

- Age band

- Urban/Rural **status**
- Device type

Major Observations

- Urban teenagers using smartphones formed the single highest screen-time cohort.
- Rural children showed comparatively lower screen-time intensity across all devices.
- Device diversity (smartphone, tablet, laptop) was significantly higher in urban localities.

This multi-dimensional view gave a deeper understanding of behavioral segmentation.

5. Cohort Ranking (Age × Device × Location)

A top-10 ranking of device-age-location combinations revealed:

- Smartphone users aged **16–18 (urban)** ranked the highest.
- Laptop usage was especially higher among **urban students aged 12–15**.
- Younger rural groups consistently remained low in screen-time ranking.

This ranking helps in identifying priority segments for digital wellness interventions.

6. Deviation Analysis (Age vs Deviation from Mean)

Purpose

To see which cohorts significantly deviate from the **overall mean screen-time**.

Insights

- Teens, especially using smartphones, show **positive deviation** from the mean.
- Young children and laptop/TV users show **negative deviation**, indicating safer screen behavior.
- The red reference line helped visually detect outlier groups with unusually high screen-time patterns.

7. Location-wise Trend Analysis (Urban vs Rural)

A dual-line plot comparing urban and rural trends across age bands showed:

- **Urban screen time increases steadily** with age.
- Rural screen time remains comparatively stable.
- The gap widens significantly in the teen age band.

This illustrates how digital environments influence behavior.

8. Quadrant Analysis (Educational Ratio vs Screen Time)

This analysis divides children into four behavioral quadrants:

- **High Screen Time + Low Educational Ratio → High-Risk Group**
- **High Screen Time + High Educational Ratio →** Balanced but intense users
- **Low Screen Time + High Educational Ratio →** Ideal/healthier users
- **Low Screen Time + Low Educational Ratio →** Recreational-low users

Conclusion

The quadrant chart visually distinguishes segments requiring digital wellness interventions, especially **teens with high recreational patterns**.

Week 6: Seasonal / Calendar / Habit Pattern Analysis

Week 6 focuses on understanding how screen-time changes over time or varies based on daily routines, seasons, or academic calendars.

1. Urban vs Rural Habit Differences

Findings

- Urban screen usage spikes more dramatically on weekends.
- Rural screen patterns remain more stable (limited device availability).
- Urban locations show stronger recreational patterns.

2. Habit Intensity Heatmap (Age Group × Day Type)

This heatmap shows:

- Teen groups have significantly higher intensity during weekends.
- Younger age groups remain consistent.

Very useful for identifying risk-oriented behavior

3. Z-score Behavior Analysis

Z-score indicates how many standard deviations a child's screen time differs from the mean.

Interpretation

- High z-score → Excessive screen users

- Low z-score → Safe/low users
- Smartphone-using teens have the highest z-scores

This highlights outliers requiring attention.

4. Peak Usage Cohort Summary

Top peak usage cohorts identified:

- Teen + Smartphone + Weekend (Urban)
- Pre-teens + Smartphone + Weekend
- Teen laptop users during weekdays (academic factor)

These provide targeted cohorts for intervention and insights

5. Demographic Habit Differences

Habit patterns differ across demographics:

- **Urban vs Rural** weekend spikes
- **Age-based changes**, where teens show stronger fluctuations
- **Gender-based differences** in activity ratios
- **Device-based stability**, where laptop users show consistent educational usage

These insights help identify which groups maintain healthy digital habits and which groups require better intervention.

6. Identifying Behavioral Drivers and Key Segments

Combining Week 5 and Week 6 results, the following drivers are identified:

- Younger children have stable, low screen-time patterns

- Teenagers show highest variability and highest recreational usage
- Smartphone-heavy users are the most at risk
- Urban children exhibit stronger weekend spikes due to more device availability.

Deliverables for Milestone 3

✓ Segment summaries

Explaining how screen-time varies by age, gender, device, and location.

✓ Heatmaps & stacked visualizations

Showing high-density usage clusters and demographic differences.

✓ Cohort insights

Identifying high-exposure cohorts such as:

- Teens using smartphones
- Urban children with laptop access
- Weekend recreational users

✓ Seasonal/Habit patterns

Observing weekly behavior and weekend spikes, or monthly trends if date data exists.

Overall Milestone 3 Conclusion

Milestone 3 provided deep behavioral insights by analyzing children's screen-time patterns through segmentation and

cohort-based exploration. The combination of Week 5 and Week 6 visual analytics revealed **which groups are most vulnerable, which devices contribute most to high usage, and how habits shift across weekdays, weekends, and seasons**. These findings enable targeted recommendations, risk assessment, and personalized digital wellness strategies.

SUBMITTED BY:

LAKSHMI REDDY KOTTAM